



ANDROID APPLICATION STATIC ANALYSIS CHEAT SHEET

Extracting and Enumerating APK Files

Command	Description
<code>adb install myapp.apk</code>	Install an application.
<code>adb shell pm path com.example.myapp</code> <code>adb shell pm list packages</code> <code>adb pull /data/app/com.example.myapp-1/base.apk .</code>	Extract the APK from the device.
<code>apktool d myapp.apk</code>	Disassembling an APK

Analyzing Application's Source Code

Command	Description
<code>sudo apt install jadx</code> <code>jadx-gui /full/path/to/myapp.apk</code>	Analyze app with JADX.
<code>curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.2/install.sh bash</code> <code>source ~/.bashrc</code> <code>nvm install node</code> <code>npm -g install js-beautify</code> <code>apktool d myapp.apk</code> <code>js-beautify myapp/assets/index.android.bundle -o beautified_index.android.bundle.js</code>	Beautify the content of the minified JS code.
<code>pip3 install --upgrade git+https://github.com/Pl1sec/hermes-dec</code> <code>hbc-decompiler index.android.bundle output.js</code>	Decompile Hermes JavaScript bytecode.

Command	Description
<pre>wget https://github.com/giacomoferretti/paranoid-deobfuscator/archive/refs/tags/v2.0.1.zip unzip v2.0.1.zip cd paranoid-deobfuscator-2.0.1 python -m venv .venv source .venv/bin/activate pip install "numpy==1.26.0" apktool d myapp.apk python -m paranoid_deobfuscator -v myapp</pre>	Deobfuscate apps obfuscated with Paranoid/LSParanoid

Analyzing Native Libraries

Command	Description
<pre>wget https://raw.githubusercontent.com/x4lsec/tools/master/Mobile/Xamarin/Xamarin_XALZ_decompress.py python Xamarin_XALZ_decompress.py myapp/unknown/assemblies/Myapp.dll myapp_decompressed.dll</pre>	Decompress special Xamarin files.
<pre>sudo pip3 install git+https://github.com/jakev/pyxamstore.git pyxamstore unpack -d myapp/unknown/assemblies/</pre>	Extract assemblies from .blob files.

Application Patching

Command	Description
<pre>adb connect 10.11.1.1:5001 adb install mybank.apk adb uninstall mynotes.apk</pre>	Connect to remote device, install and uninstall apps.
<pre>apktool b myapp echo -e "password\npassword\njohn doe\ntest\ntest\ntest\ntest\ntest\nyes" > params.txt cat params.txt keytool -genkey -keystore key.keystore -validity 1000 -keyalg RSA -alias john zipalign -p -f -v 4 myapp/dist/myapp.apk myapp_aligned.apk echo password apksigner sign --ks key.keystore myapp_aligned.apk adb uninstall com.hackthebox.myapp adb install myapp_aligned.apk</pre>	Recompile an APK, sign it, uninstall the old one, and install the patched one.